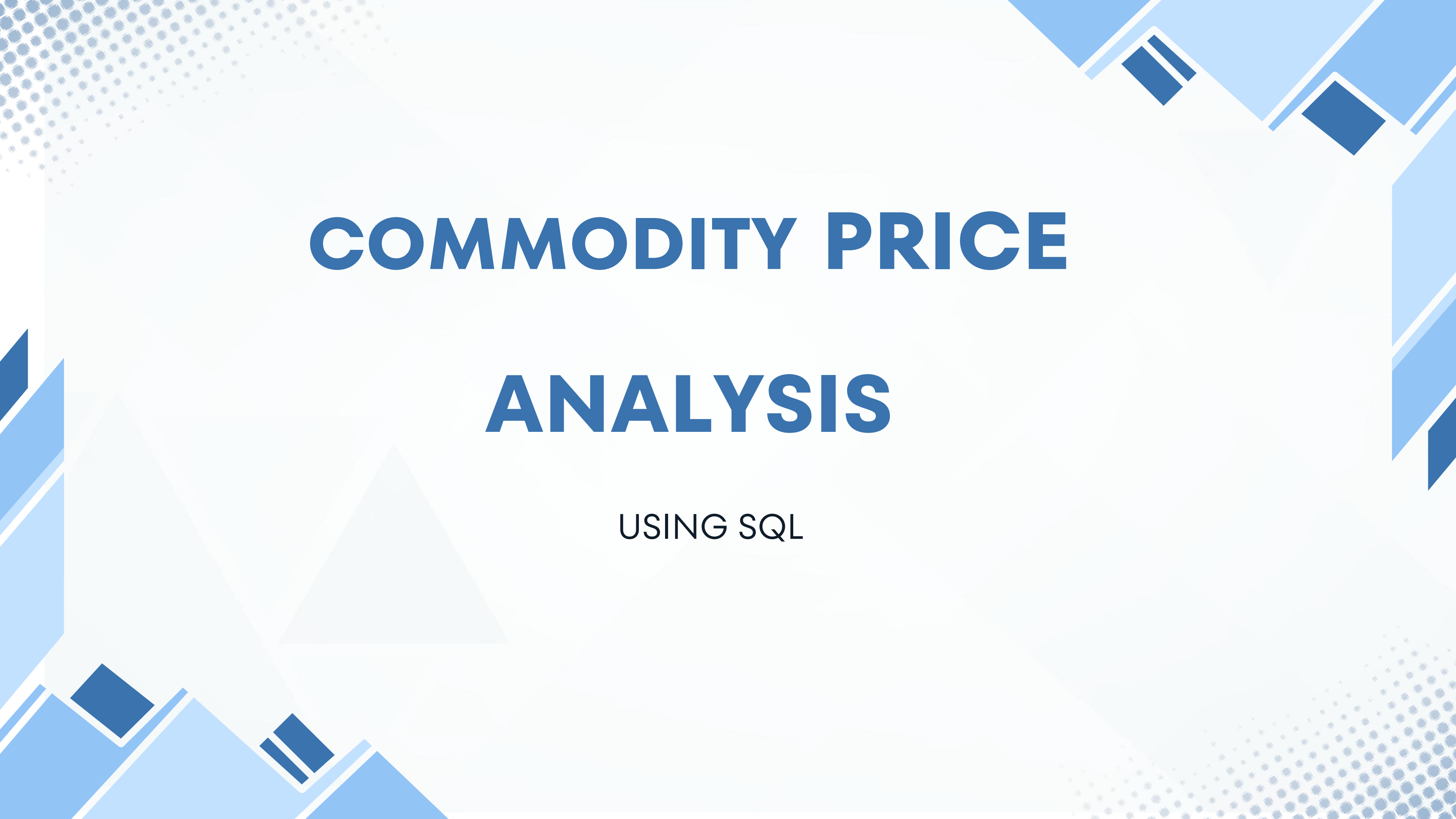




COMMODITY PRICE ANALYSIS

USING SQL

PROJECT DESCRIPTIONS

This project focuses on analyzing commodity price data using SQL to identify price trends, regional variations, and commodity availability across different states and cities. The analysis was performed using multiple datasets related to commodities, regions, and retail prices.

Dataset Overview

The project uses three datasets:

1. price_details – Contains commodity IDS, region IDs, dates, and retail prices
2. commodities_info – Contains commodity IDs, names, varieties, and categories
3. region_info – Contains region IDs, state and city information

Technology Used

- MySQL Workbench
- PowerPoint for reporting

Data Cleaning

- Checked for null values
- Checked duplicate records
- Verified data types
- Identified blank spaces in the **Variety** column(commodities_info)
- Replaced blank variety values with 'Unknown'

SQL Concepts Used

- JOINS
- GROUP BY
- ORDER BY
- Aggregate Functions (COUNT, MAX, MIN)
- CTE (Common Table Expressions)
- Filtering using WHERE and HAVING
- LIMIT and OFFSET

Business Questions Solved

1. Identified common commodities among the Top 10 costliest commodities in 2019 and 2020

Query

```
with costliest_2019 as(  
  select c.Commodity from price_details p  
  join commodities_info c  
  on p.Commodity_Id=c.Id  
  where year(p.Date)=2019  
  group by c.Commodity  
  order by max(p.Retail_Price) desc  
  limit 10  
,  
costliest_2020 as(  
  select c.Commodity from price_details p  
  join Commodities_info c  
  on p.Commodity_Id=c.Id  
  where year(p.Date)=2020  
  group by c.Commodity  
  order by max(p.Retail_Price) desc  
  limit 10  
)  
  
select y19.Commodity as Common_Commodity  
from costliest_2019 y19  
inner join costliest_2020 y20  
on y19.Commodity=y20.Commodity;
```

Output

commodity
Shoes-Gents
Black Pepper
Coffee
Soft Cake
Meat
Ghee
Saree
Sandals-Ladies
Cement

Business Insight: High-value commodities like Gents Shoes, Black Pepper, Coffee, Meat, and Ghee consistently maintain their premium pricing status across multiple years (2019 and 2020). Demand or production costs for these goods remain structurally high, unaffected by yearly market fluctuations.

Business Questions Solved

2. Found the commodity with the maximum price difference across regions in June 2020

Query

```
WITH june_data AS (  
  SELECT  
    c.commodity,  
    MIN(p.retail_price) AS min_price,  
    MAX(p.retail_price) AS max_price  
  FROM price_details p  
  JOIN commodities_info c  
    ON p.commodity_id = c.id  
  WHERE MONTH(p.date) = 6  
    AND YEAR(p.date) = 2020  
  GROUP BY c.commodity  
)  
  
SELECT  
  commodity,  
  (max_price - min_price) AS price_difference  
FROM june_data  
ORDER BY price_difference DESC  
LIMIT 1;
```

Output

Result Grid		Filter Rows:
	commodity	price_difference
►	Coffee	2435.00

Business Insight Coffee displays extreme geographic price volatility, showing a massive price difference of 2,435.00 between different locations within the exact same month (June 2020). This points to severe local supply chain bottlenecks, regional heavy taxation, or highly localized demand spikes.

Business Questions Solved

3.Arrange the commodities in order based on the number of varieties in which they are available, with the highest one shown at the top. Which is the 3rd commodity in the list?

Query

```
WITH variety_count AS (  
  
    SELECT  
        commodity,  
        COUNT(DISTINCT variety) AS variety_count  
    FROM commodities_info  
    GROUP BY commodity  
)  
  
SELECT  
    commodity,  
    variety_count  
FROM variety_count  
ORDER BY variety_count DESC  
LIMIT 1 OFFSET 2;
```

Output

Result Grid		Filter Rows:
	commodity	variety_count
▶	Coffee	2

Business Insight: Coffee ranks as the 3rd most widely available commodity by variety, featuring 2 distinct product variations across the market. This indicates a moderately diversified product portfolio that caters well to differing consumer preferences.

Business Questions Solved

4. In the state with the least number of data points available. Which commodity has the highest number of data points available?

Query

```
WITH state_count AS (  
  SELECT r.state,  
         COUNT(p.id) AS total_records  
  FROM price_details p  
  JOIN region_info r  
    ON p.region_id = r.id  
  GROUP BY r.state  
  ORDER BY total_records ASC  
  LIMIT 1  
,  
commodity_count AS (  
  SELECT c.commodity,  
         COUNT(p.id) AS commodity_records  
  FROM price_details p  
  JOIN region_info r  
    ON p.region_id = r.id  
  JOIN commodities_info c  
    ON p.commodity_id = c.id  
  WHERE r.state = (  
    SELECT state  
    FROM state_count  
  )  
  GROUP BY c.commodity  
)  
SELECT commodity  
FROM commodity_count  
ORDER BY commodity_records DESC  
LIMIT 1;
```

Output

Result Grid		Filter Rows:
	commodity	
▶	Rice	

Business Insight: Even in the state with the sparsest overall data coverage, Rice emerges as the dominant commodity with the highest volume of record entries. Rice remains the staple market driver and most consistently tracked item even where regional data collection infrastructure is weak.

Business Questions Solved

5. What is the price variation of commodities for each city from Jan 2019 to Dec 2020. Which commodity has seen the highest price variation and in which city?

Query

```
WITH jan_2019 AS (  
  SELECT Region_Id, Commodity_Id, Retail_Price AS Start_Price  
  FROM price_details  
  WHERE Date LIKE '2019-01%' ),  
dec_2020 AS (  
  SELECT Region_Id, Commodity_Id, Retail_Price AS End_Price  
  FROM price_details  
  WHERE Date LIKE '2020-12%' )  
SELECT  
  c.Commodity,  
  r.Centre,  
  j.Start_Price AS 'Start Price',  
  d.End_Price AS 'End Price',  
  (d.End_Price - j.Start_Price) AS 'Variation Absolute',  
  (((d.End_Price - j.Start_Price) / j.Start_Price) * 100) AS 'Variation Percentage'  
FROM jan_2019 j  
INNER JOIN dec_2020 d ON j.Region_Id = d.Region_Id AND j.Commodity_Id = d.Commodity_Id  
JOIN region_info r ON j.Region_Id = r.Id  
JOIN commodities_info c ON j.Commodity_Id = c.Id  
ORDER BY 'Variation Percentage' DESC  
LIMIT 1;
```

Output

Result Grid Filter Rows: Export: Wrap Cell Content:						
	Commodity	Centre	Start Price	End Price	Variation Absolute	Variation Percentage
▶	Moong	Chittoor	90.00	110.00	20.00	22.222222

Business Insight: This query identifies the ultimate macro-level inflation or supply shock outlier across a two-year horizon.

Key Insights

- Some commodities remained consistently expensive across multiple years.
- Certain commodities showed high regional price variation.
- Rice had one of the highest variety counts.
- Price fluctuations varied significantly across cities and states.
- Commodity trends helped identify regional market differences.

Challenges faced

- Understanding multi-table joins
- Handling aggregation and grouping logic
- Working with date-based filtering
- Managing blank values and data cleaning
- Converting business problems into SQL queries

Conclusion

This project demonstrated how SQL can be used to analyze real-world commodity pricing data and generate meaningful business insights. The analysis improved understanding of SQL concepts, data handling, and problem-solving using relational databases.



SUPPLY CHAIN DATA ANALYSIS

USING SQL

PROJECT DESCRIPTIONS

This project focuses on analyzing supply chain and order management data using SQL to identify customer behavior, shipment performance, sales trends, and operational efficiency. The analysis was performed using multiple datasets related to orders, customers, products, departments, and shipments.

Dataset Overview

The project uses multiple datasets:

1. orders – Contains order details, shipment details, transaction type, and order status etc.
2. ordered_items – Contains product sales information
3. customer_info – Contains customer details such as city and state
4. product_info – Contains product and department details
5. department – Contains department names

Technology Used

- MySQL Workbench
- PowerPoint for reporting

Data Cleaning

- Checked for null values
- Checked duplicate records
- Verified data types
- Identified **blank spaces** in the **Last_Name** **column(customer_info)**
- Replaced blank variety values with 'Unknown'
- Corrected inconsistent state names (Orders)
- (e.g. "Ammu and Kashmir" → "Jammu and Kashmir")

SQL Concepts Used

- JOINS
- GROUP BY
- ORDER BY
- Aggregate Functions (COUNT, SUM)
- CASE Statements
- CTE (Common Table Expressions)
- HAVING Clause
- Filtering using WHERE conditions

Business Questions Solved

1: Get the number of orders by the Type of Transaction. Please exclude orders shipped from Sangli and Srinagar. Also, exclude the SUSPECTED_FRAUD cases based on the Order Status. Sort the result in the descending order based on the number of orders.

Query

```
SELECT
  Type AS 'Type of Transaction', COUNT(Order_Id) AS Orders
FROM
  orders
WHERE
  Order_City NOT IN ('Sangli' , 'Srinagar')
  AND Order_Status <> 'SUSPECTED_FRAUD'
GROUP BY Type
ORDER BY Orders DESC;
```

Output

Result Grid			Filter Rows:
	Type of Transaction	Orders	
▶	DEBIT	799	
	TRANSFER	532	
	PAYMENT	488	
	CASH	259	

Business Insight: Customer Payment Preferences ,Targeted Market Analysis
Risk-Adjusted Volume

Business Questions Solved

2.Get the list of the Top 3 customers based on the completed orders along with the following details: Customer Id, Customer First Name, Customer City, Customer State, Number of completed orders, Total Sales

Query

```
SELECT
  c.Id AS 'Customer Id',
  c.First_Name AS 'Customer First Name',
  c.City AS 'Customer City',
  c.State AS 'Customer State',
  COUNT(DISTINCT o.Order_Id) AS 'Completed orders',
  ROUND(SUM(oi.Sales), 2) AS 'Total Sales'
FROM
  orders o
JOIN ordered_items oi ON o.Order_Id = oi.Order_Id
JOIN customer_info c ON o.Customer_Id = c.Id
WHERE
  o.Order_Status = 'COMPLETE'
GROUP BY
  c.Id, c.First_Name, c.City, c.State
ORDER BY
  'Completed orders' DESC,
  'Total Sales' DESC
LIMIT 3;
```

Output

	Customer Id	Customer First Name	Customer City	Customer State	Completed orders	Total Sales
▶	5612	Mary	Caguas	PR	3	2188.82
	3465	Mary	Santa Clara	CA	2	2929.74
	7217	Carl	Caguas	PR	2	1239.83

Business Insight:

Identification of VIP/Power Users: It highlights the most loyal, high-frequency customers.

Targeted Customer Retention & Perks: Knowing these specific IDs allows marketing teams to enroll them into VIP loyalty programs, target them with exclusive high-tier rewards, or cross-sell premium products, ensuring they don't churn to competitors.

Business Questions Solved

3.Get the order count by the Shipping Mode and the Department Name. Consider departments with at least 40 closed/completed orders.

Query

```
387 -- Final Output
388
389 • select d.name as Department_Name,o.Shipping_Mode, count(o.Order_Id) as Total_Orders
390 from orders o
391 join ordered_items oi
392 on o.Order_Id=oi.Order_id
393
394 join product_info p
395 on oi.Item_Id = p.Product_Id
396
397 join department d
398 on p.Department_Id=d.Id
399 where o.Order_Status in ('COMPLETE', 'CLOSED')
400 group by d.Name,o.Shipping_Mode
401 having COUNT(o.order_id)>=40 ;
402
403
```

Output

Result Grid				Filter Rows:	Search	Export:
	Department_Na...	Shipping_Mode	Total_Orders			
	Fan Shop	Second Class	165			
	Golf	Second Class	75			
	Footwear	Standard Class	86			
	Fan Shop	Standard Class	371			
	Outdoors	Standard Class	73			
	Golf	Standard Class	186			
	Apparel	First Class	84			
	Fan Shop	First Class	119			
	Apparel	Standard Class	300			
	Apparel	Second Class	143			
	Golf	First Class	57			
	Discs Shop	Standard Class	85			

Business Insight:

The analysis of shipping modes and department-wise completed orders showed that some departments consistently received a higher number of successful orders, indicating strong customer demand and better product performance.

These insights can help improve inventory management, shipment planning, and overall supply chain performance.

Business Questions Solved

4.reate a new field as shipment compliance based on Real_Shipping_Days and Scheduled_Shipping_Days.

It should have the following values:

Cancelled shipment - If the Order Status is SUSPECTED_FRAUD or CANCELED

Within schedule - If shipped within the scheduled number of days

On time - If shipped exactly as per schedule

Upto 2 days of delay - If shipped beyond schedule but delay upto 2 days

Beyond 2 days of delay - If shipped beyond schedule with delay beyond 2 days

Output

Result Grid					Filter Rows:		Search	Export:
Order_Id	Shipment_Complian...	Shipping_Mo...	Number_delayed_ord...					
72520	Upto 2 days of delay	Standard Class	466					

```
Query
Limit to 1000 rows

501 -- Final output
502
503 • select Order_Id,shipment_compliance,Shipping_Mode,
504 count(Order_Id) over (partition by Shipping_Mode) as Number_delayed_orders
505 from (
506   select Order_Id,Shipping_Mode,
507   case
508   when Order_Status in ('SUSPECTED_FRAUD','CANCELED')
509   then 'Cancelled Shipment'
510   when Scheduled_Shipping_Days > Real_Shipping_Days
511   then 'Within Schedule'
512   when Real_Shipping_Days = Scheduled_Shipping_Days
513   then 'On time'
514   when Real_Shipping_Days > Scheduled_Shipping_Days +2
515   then 'Beyond 2 days of delay'
516   when Real_Shipping_Days > Scheduled_Shipping_Days
517   and Real_Shipping_Days<=Scheduled_Shipping_Days+2
518   then 'Upto 2 days of delay'
519   Else 'Unknown'
520   end as Shipment_Compliance
521   from orders
522 ) F
523 where shipment_compliance IN ('Upto 2 days of delay','Beyond 2 days of delay')
524 order by Number_delayed_orders desc
525 limit 1;
```

Business Insight:

The shipment compliance analysis helped classify orders based on delivery performance, such as on-time deliveries, delayed shipments, and cancelled shipments.

Business Questions Solved

5. An order is cancelled when the status of the order is either CANCELED or SUSPECTED_FRAUD. Obtain the list of states by the order cancellation % and sort them in the descending order.

Query

```
628 -- final output
629
630 with cancelled as
631 (
632   select Order_State, count(Order_Id) as CANCELED_orders
633   from Orders
634   where order_Status in ('CANCELED', 'SUSPECTED_FRAUD')
635   group by Order_State ),
636
637 total as
638 (
639   select Order_State, count(Order_Id) Total_orders
640   from Orders
641   group by Order_State
642 )
643
644 select c.Order_State,
645        round((c.CANCELED_orders /t.Total_orders)*100,2 )as Cancellation_Percentage
646
647 from cancelled c |
648 join total t
649 on c.Order_state=t.Order_State
650 order by Cancellation_Percentage desc;
651
652
```

Output

Result Grid		Filter Rows:	Search
Order_State	Cancellation_Percenta...		
Manipur	11.11		
Punjab	9.09		
Haryana	9.09		
Jharkhand	8.82		
Puducherry	8.33		
Uttar Pradesh	6.25		
Telangana	5.88		
Andhra Pradesh	5.21		
Jammu and Kashmir	4.55		
Kerala	4.35		
Delhi	4.11		
Karnataka	4.09		
Madhya Pradesh	3.77		
Odisha	3.45		
Chhattisgarh	2.94		
Maharashtra	2.80		
Tamil Nadu	2.55		
Bihar	2.30		
West Bengal	1.64		
Gujarat	1.63		
Rajasthan	0.67		

Business Insight:

The cancellation percentage analysis revealed that some states experienced significantly higher order cancellation rates compared to others.

Key Insights

- Certain transaction types contributed significantly more orders.
- Top customers generated high completed orders and sales revenue.
- Some departments showed consistently high order volumes.
- Standard shipping modes experienced higher shipment delays.
- Cancellation percentages varied across different states.

Challenges faced

- Handling multi-table joins
- Building shipment compliance logic using CASE statements
- Working with aggregate filtering using HAVING
- Managing order status conditions and delay calculations
- Translating business requirements into SQL queries

Conclusion

This project demonstrated how SQL can be used to analyze supply chain operations, shipment performance, customer behavior, and order trends. The analysis improved understanding of SQL querying, data handling, and business problem-solving using relational databases.



THANK YOU

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